# Client–Server Chat Application

This project consists of designing and implementing a real-time client–server chat application using C# and .NET. The system will rely on TCP socket communication and support multiple simultaneous clients through concurrent programming techniques.

The server will manage connections, user sessions, and message distribution. **Message broadcasting will be implemented using C# delegates and events**, allowing a decoupled and extensible architecture for notifying connected clients.

## Core Functionalities

### Server-Side Functions

1. Start and stop the server.
2. Accept multiple client connections via TCP sockets.
3. Authenticate users (username-based login).
4. Maintain a list of connected clients.
5. Receive messages from clients.
6. Broadcast messages to all connected clients using delegates.
7. Send private messages to specific users.
8. Log server activity (connections, disconnections, errors).

You can extend this list as you wish (multiple chatrooms, message logging, ...)

### Client-Side Functions

1. Connect to the server.
2. Send login credentials or username.
3. Send public messages.
4. Receive and display broadcast messages.
5. Send and receive private messages.

### Basic Commands (optional but recommended)

- /name to change username
- /quit to leave the chat
- /users to list connected users

### Technical Requirements

1. Use a client–server architecture.
2. Use sockets for communication.

3. The server must support multiple clients simultaneously (threads or asynchronous programming).
4. Messages should be transmitted in real time **using delegates**.

## Evaluation

During the final session, you will be evaluated and asked questions about your work as well as your understanding of the different concepts we have covered.

## Deliverables (usually 1 or 2 weeks after the evaluation)

Students must submit :

1. The source code for the server and client.
2. A short documentation explaining
    a. The architecture
    b. The important parts of the implementation
    c. Screenshots of the application and tests